

4. (a) A lithium **atom** is denoted by  ${}^7_3\text{Li}$ .

**Complete** the table for the lithium atom. One cell has already been completed for you.

[3]

| Particles   | Number in ${}^7_3\text{Li}$ atom |
|-------------|----------------------------------|
| electrons   | 3                                |
| protons     |                                  |
| neutrons    |                                  |
| leptons     |                                  |
| baryons     |                                  |
| up quarks   |                                  |
| down quarks |                                  |

- (b) (i) State what a meson is, in terms of quarks.

[1]

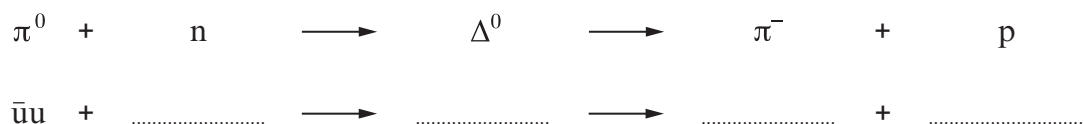
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- (ii) Explain why the quark make-up of a  $\pi^0$  meson must be either  $u\bar{u}$  or  $d\bar{d}$ .

[1]

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- (iii) The  $\Delta^0$  particle is a baryon which has a charge equal to that of a neutron. The following series shows the production **and** decay reactions of the  $\Delta^0$  particle. The reactions are also partially shown as a flow of quarks.



**Complete** the reactions in terms of quarks. The  $\pi^0$  has been done for you.

[4]



- (iv) Explain clearly how the up quark number is conserved in both reactions.  
[Consider both reactions separately.] [2]

Production of  $\Delta^0$  : .....

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Decay of  $\Delta^0$  : .....

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- (c) Textbooks state that the mean lifetime of the  $\Delta^0$  particle is  $(5.63 \pm 0.14) \times 10^{-24}$  s.  
State **two** features of the decay which suggest that it is a **strong** interaction. [2]

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